

OSI Model

OSI (Open Systems Interconnection) Reference Model is a conceptual framework developed by the International Organization for Standardization (ISO) to standardize and describe how network communication functions are organized.

- The OSI model divides network communication into **seven layers**, with each layer responsible for a specific group of networking functions.
- The OSI model is **not itself a networking protocol**. Instead, it provides a common framework for understanding, designing, comparing, and troubleshooting network protocols and technologies.

OSI Layering Concepts and Benefits

Networking involves many complex functions. Dividing these functions into smaller groups, called **layers**, provides several important benefits:

- Reduces complexity by dividing networking tasks into manageable parts.
- Provides modularity, allowing each layer to perform a specific set of functions.
- Simplifies troubleshooting by helping identify the layer where a problem occurs.
- Facilitates protocol development by defining functional responsibilities for each layer.
- Promotes interoperability between different networking technologies and vendors.
- Allows changes in one layer without necessarily requiring changes to all other layers.
- Provides a standard reference framework for understanding and comparing networking protocols.

Today, the OSI model is primarily used as a **reference and teaching model** for understanding networking concepts and comparing different networking architectures.

For example, **IP** is commonly described as a **Layer 3 (Network layer) protocol** because its primary functions correspond to the OSI Network layer.

Notes on the OSI Model:

Upper Layers

The upper three layers—Application, Presentation, and Session (Layers 7, 6, and 5)—are concerned primarily with functions that support applications and communication between application processes. They are sometimes collectively referred to as the **Application layers**.

Lower Layers

The lower four layers—Transport, Network, Data Link, and Physical (Layers 4, 3, 2, and 1)—are primarily concerned with the transport and delivery of data across a network. They are commonly referred to as the **Data Flow Layers**.

Note: The terms "**Application Layers**" and "**Data Flow Layers**" are not standard OSI terminology. For teaching purposes, **upper layers** and **lower layers** are preferable.

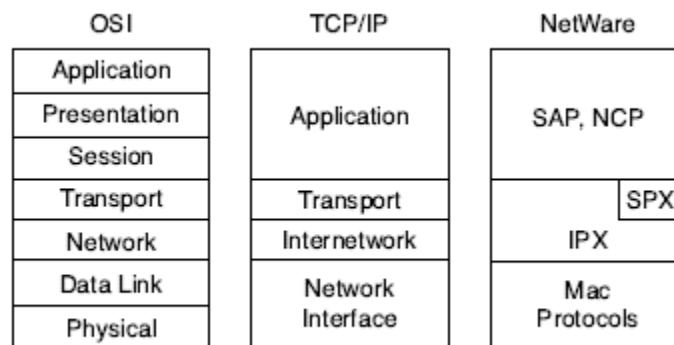
General Principles

- Each layer performs a **specific set of functions**.
- Each layer provides services to the layer above it.
- Each layer uses the services provided by the layer below it.
- A layer communicates logically with its **peer layer** on another device.
- During transmission, data is **encapsulated** as it moves down the layers.
- At the receiving device, data is **decapsulated** as it moves up the layers.

OSI Layers:

The OSI model contains seven layers:

1. Physical Layer – Layer 1
2. Data Link Layer – Layer 2
3. Network Layer – Layer 3
4. Transport Layer – Layer 4
5. Session Layer – Layer 5
6. Presentation Layer – Layer 6
7. Application Layer – Layer 7



Main Functions of the OSI Layers

Layer 7 – Application

The **Application layer** provides network-related services directly to applications.

For example, when a web browser requests a web page, application-layer protocols provide the mechanisms required for communication with the web server.

Examples: HTTP, FTP, SMTP, DNS, SNMP

Layer 6 – Presentation

The **Presentation layer** is concerned with the **representation and format of data** so that communicating systems can interpret the information consistently. Typical functions include:

- Data format translation
 - Character encoding
 - Encryption/decryption
 - Compression/decompression
- Examples/technologies traditionally associated with this layer: ASCII, EBCDIC, JPEG, MPEG

Layer 5 – Session

The **Session layer** manages communication sessions between applications. Its functions may include:

- Establishing a session
 - Maintaining a session
 - Synchronizing communication
 - Managing dialogs or transactions
 - Terminating a session
- Examples of technologies historically associated with Session-layer functions include **RPC** and some session-management mechanisms.

Layer 4 – Transport

The **Transport layer** provides **end-to-end communication** between applications running on different devices. Its major functions include:

- Segmentation and reassembly
- End-to-end delivery
- Reliability when supported by the protocol
- Flow control
- Error recovery when supported

- Multiplexing using port numbers

Examples: TCP, UDP

TCP provides reliable, connection-oriented delivery, while UDP provides connectionless delivery without TCP's built-in reliability mechanisms.

Layer 3 – Network

The Network layer is responsible primarily for delivering packets between different networks. Its major functions include:

- Logical addressing
- Routing / path determination

Example: IP

For this reason, IP is commonly referred to as a **Layer 3 protocol**.

Layer 2 – Data Link

The **Data Link layer** provides communication between directly connected network nodes over a particular link or network segment. Its major functions include:

- Framing
- MAC addressing
- Media access control
- Error detection
- Link-level flow control in some technologies

A Data Link frame normally contains: Header + Network-layer Packet + Trailer

The header may contain source and destination MAC addresses, while the trailer commonly contains an FCS (Frame Check Sequence) for error detection.

Examples: Ethernet (IEEE 802.3), IEEE 802.2 LLC, HDLC

Layer 1 – Physical

The Physical layer is responsible for transmitting raw bits over the physical medium. It defines characteristics such as:

- Electrical or optical signals
- Voltage levels

- Signaling methods
- Bit rate
- Connectors
- Cable specifications
- Pin assignments
- Physical transmission media

Examples:

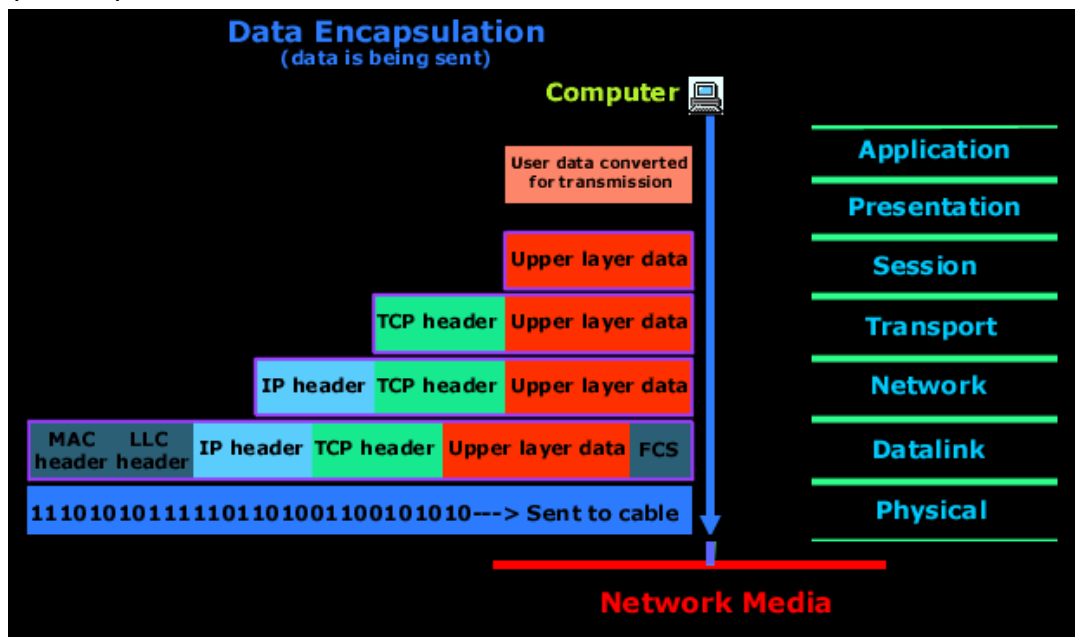
- EIA/TIA-232
- V.35
- Physical Ethernet specifications
- Fiber-optic transmission standards

OSI Reference Model Layer Definitions and Functional Summary:

Layer #	Layer Name	Definition & Function	Example
7	Application	Interface between network and application software (Used by User).	Telnet, HTTP,FTP
6	Presentation	How data is presented. It performs special processing, such as encryption. Compression & Coding of data	JPEG,ASCII, EBCDIC
5	Session	Establishing and maintaining end-to-end bidirectional flows between endpoints. Includes managing transaction flows.	Operating systems and application access scheduling RPC
4	Transport	It provides Reliable or unreliable delivery. It provides Multiplexing.	TCP, UDP, SPX
3	Network	Logical addressing, Routing (path determination).	IP, IPX
2	Data Link	Encapsulation (Framing) :Combination of bits into bytes Access to the media using MAC address. Error detection and Error recovery.	802.3/802.2, HDLC
1	Physical	Moving of bits between devices. It Provides Specification of voltage, wire speed, and cable pinouts.	EIA/TIA-232, V.35

Encapsulation is the process of adding protocol-specific control information to data as it moves down through the layers of the OSI or TCP/IP model on the sender's side.

- This additional information is mainly added in the form of headers, and at the Data Link layer, a trailer may also be added. Encapsulation occurs as data moves from the upper layers toward the Physical layer.



Working of Encapsulation

- Application Layer**

The user creates data, such as an email, web request, or message. The Application layer provides the data to the lower layers.

- Presentation and Session layers** may also process the data before it reaches the Transport layer. These layers do not normally add a header or trailer in the same way that the Transport, Network, and Data Link layers do.

- Transport Layer**

The Transport layer may divide the application data into smaller units called segments.

It adds a Transport-layer header containing information such as:

- ✓ Source port number
- ✓ Destination port number
- ✓ Sequence number
- ✓ Acknowledgment information
- ✓ Other control information

If TCP is used, the resulting unit is called a TCP segment. If UDP is used, the resulting unit is called a UDP datagram.

- **Network Layer**

The Transport-layer data is passed to the Network layer. The Network layer adds an IP header, which contains information such as:

- ✓ Source IP address
- ✓ Destination IP address
- ✓ Other network-layer control information

The resulting unit is called an IP packet.

- **Data Link Layer**

The Network-layer packet is passed to the Data Link layer. The Data Link layer adds a header and usually a trailer to form a frame. The header may contain information such as:

- ✓ Source MAC address
- ✓ Destination MAC address
- ✓ Frame-control information
- ✓ The trailer commonly contains an FCS (Frame Check Sequence) used for error detection.

The resulting unit is called a Frame.

- **Physical Layer**

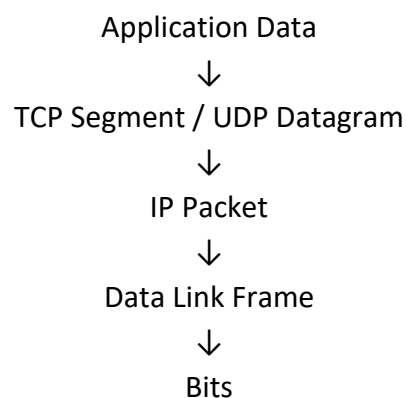
The Data Link layer passes the frame to the Physical layer. The Physical layer converts the frame into a sequence of bits and transmits those bits through the physical transmission medium.

Encapsulation Process

The encapsulation process can be summarized as:

Data → Segment/Datagram → Packet → Frame → Bits

Or, more specifically:



- At the receiving side, the reverse process occurs. This is called decapsulation, where each layer removes and processes its corresponding header or trailer before passing the remaining data to the next upper layer.

A **Protocol Data Unit (PDU)** is the unit of data exchanged between network layers according to the protocol operating at that layer.

As data moves down through the layers during **encapsulation**, it is given different names at different layers. These names identify the PDU associated with each layer.

Application	Data
Presentation	Data
Session	Data
Transport	Segments/ Datagrams
Network	Packets
Data Link	Frames
Physical	Bits

De-encapsulation

De-encapsulation is the reverse process of encapsulation. It occurs at the receiving device as data moves upward through the protocol stack, from the Physical layer toward the Application layer.

During de-encapsulation, each layer receives the PDU from the layer below, processes the information associated with its own layer, and removes the corresponding protocol header or trailer when appropriate. The remaining data is then passed to the next upper layer.

Working of De-encapsulation

1. Physical Layer

- The Physical layer receives the transmitted bits from the physical transmission medium.
- It converts the received signals into a sequence of bits and passes the bits to the Data Link layer.

At this point: **Bits → Frame**

2. Data Link Layer

- The Data Link layer receives the frame.
- It processes the Layer 2 information, including the destination MAC address, and performs error detection using information such as the Frame Check Sequence (FCS).
- If the frame is valid and is intended for the receiving device, the Data Link layer removes the Layer 2 header and trailer.
- The remaining data is the Network-layer packet, which is passed to the Network layer.

Frame → Packet

3. Network Layer

- The Network layer receives the packet.
- It processes the IP header, including the destination IP address, and determines whether the packet should be delivered to the local device.
- After processing the Network-layer information, the IP header is removed and the remaining Transport-layer PDU is passed to the Transport layer.

Packet → Segment/Datagram

4. Transport Layer

- The Transport layer receives the TCP segment or UDP datagram.
- It processes the Transport-layer header, including information such as:
 - Source port
 - Destination port
 - Sequence and acknowledgment information in TCP
 - Other transport-control information

The Transport layer removes its header and delivers the resulting application data to the appropriate application process.

Segment/Datagram → Data

5. Upper Layers

The resulting data is passed through the **Session, Presentation, and Application layers** as applicable. These layers may perform functions such as:

- Managing the communication session
- Decrypting data
- Decompressing data
- Converting data formats
- Delivering the data to the appropriate application

Finally, the application receives the original data.

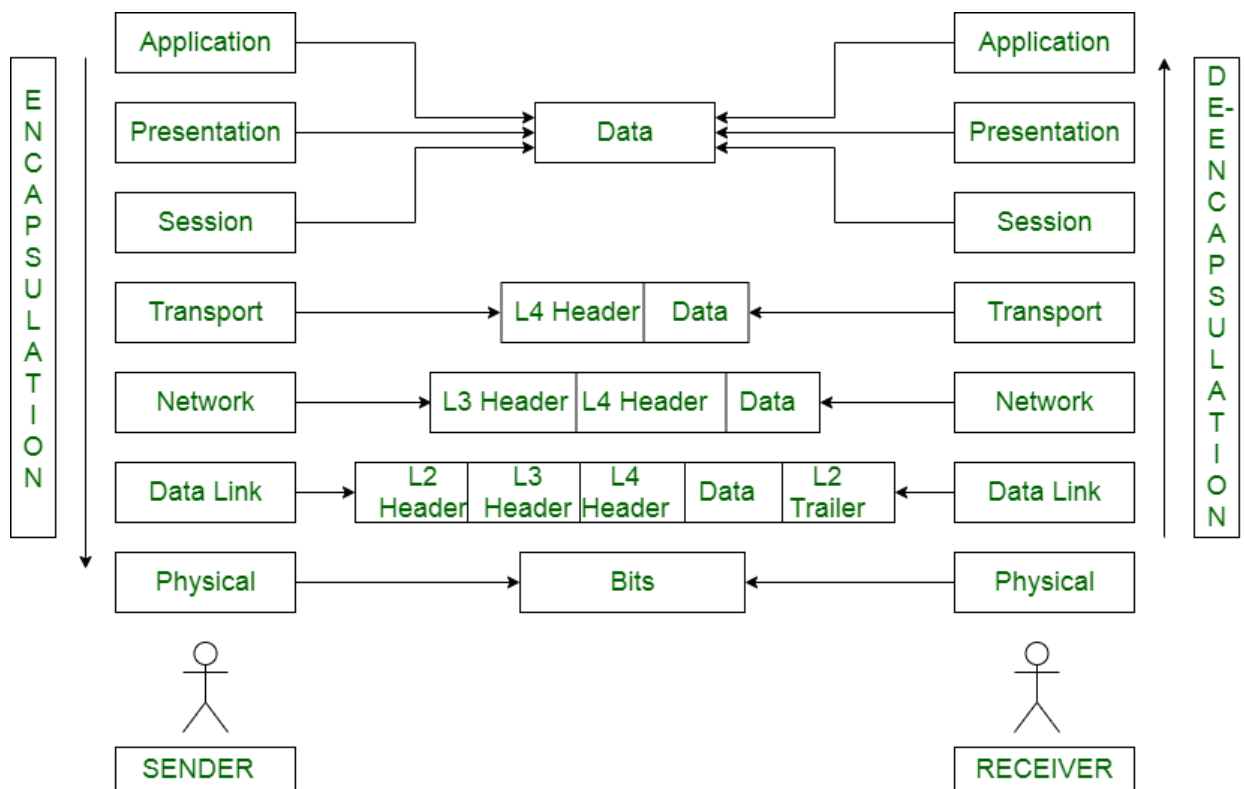
De-encapsulation Sequence: Bits → Frame → Packet → Segment/Datagram → Data

Therefore:

Encapsulation: Data → Segment/Datagram → Packet → Frame → Bits

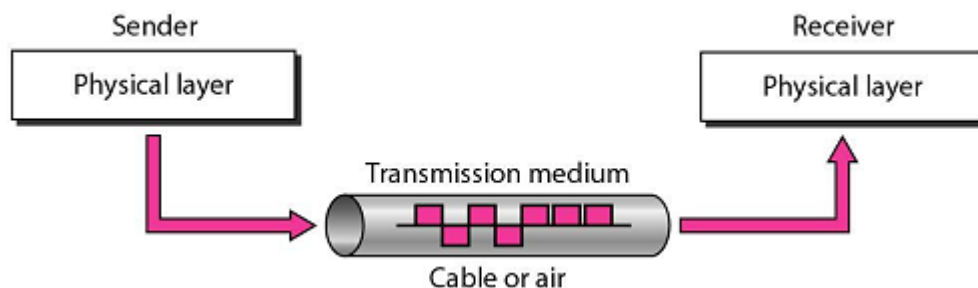
De-encapsulation: Bits → Frame → Packet → Segment/Datagram → Data

Process	Direction	Main Idea
Encapsulation	Upper → Lower	Headers/trailers are added
De-encapsulation	Lower → Upper	Headers/trailers are processed and removed



Physical Layer – Layer 1

- The Physical Layer is the first and lowest layer of the OSI (Open Systems Interconnection) model.
- It is responsible for transmitting raw bits (0s and 1s) from one device to another through a physical transmission medium.
- It is responsible for interacting with the physical transmission medium, hardware interfaces, and signaling mechanisms used to transmit data between network devices.
- The Physical Layer does not understand the meaning of the data. It only deals with how bits are represented, transmitted, and received as signals.

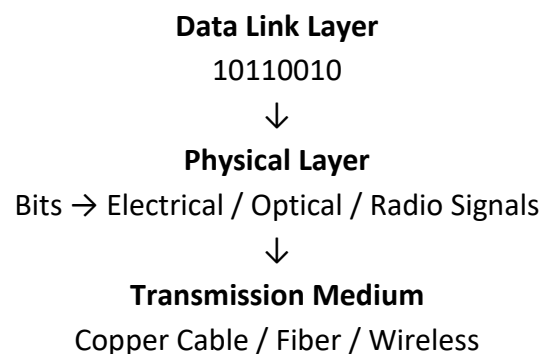


- It provides the connection between the **Data Link Layer** and the **physical transmission medium**.

Simple idea:

The Data Link Layer gives the Physical Layer a sequence of bits, and the Physical Layer converts these bits into signals that can travel through the medium.

For example:



Function	Description
Bit Transmission	Transmits raw bits between devices
Signal Representation	Represents bits using electrical, optical, or radio signals
Encoding	Defines how bits are represented by signal patterns
Modulation	Modifies a carrier signal to carry information
Transmission Medium	Defines characteristics of the physical communication medium
Data Rate	Defines the rate at which bits are transmitted
Synchronization	Helps sender and receiver identify bit boundaries
Physical Interfaces	Defines physical connection characteristics
Connectors	Specifies physical connection interfaces
Transmission Modes	Supports simplex, half-duplex, and full-duplex communication
Physical Topology	Describes physical arrangement of connections
Signal Quality	Addresses physical signal characteristics and impairments

Examples of technologies involving Physical Layer specifications include:

- Ethernet
- Wi-Fi
- Fiber-optic Ethernet
- DSL
- Cellular communication

Organizations involved in networking standards include:

- IEEE
- ITU-T
- ISO
- ANSI

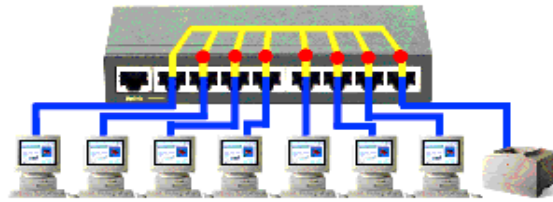
Hub

A Hub is a basic networking device that operates at Layer 1 (Physical Layer) of the OSI model. It is essentially a multi-port repeater.

Main Characteristics of a Hub

- A hub is a multi-port repeater that receives a signal on one port and regenerates and forwards the signal to all other ports.
- A hub operates at the Physical Layer (Layer 1) of the OSI model. Therefore, it does not examine or understand MAC addresses, IP addresses, frames, or packets.
- When a hub receives a transmission from one device, it repeats the signal to all connected devices, regardless of which device is the intended destination.

Hub



Security Consideration

Because a hub sends the signal to all connected ports, traffic transmitted through a hub can potentially be observed by other connected devices.

This makes a hub less private and less secure than a modern switched network.

However, it is better to say: "A hub provides no traffic isolation between its ports."

A **collision domain** : is a part of a network in which devices share the same communication medium and where two or more devices transmitting at the same time can cause their signals to collide.

When a collision occurs, the transmitted data may be corrupted and must be retransmitted.

Collision Domain of HUB

All devices connected to a hub share **one collision domain**.

This means that if two devices transmit at the same time, their signals may interfere with each other, resulting in a collision.